



Fluke 233 digital multimeter (Courtesy: TTL Technologies)

off-the-shelf (COTS) bench equipment available that fit the general requirements. However, "the difference may lie in the frequency ranges of the applications and implementation," says Jain. Illustrating with an example, he says, "The radars used in aerospace and defence typically operate below 40 GHz and involve complex design and stringent tests. However, radars used in automobiles for collision prevention work in the 76-77GHz band with a relatively simpler set-up. The testing solutions vary to match these specifications."

Digital and mixed-signal oscilloscopes, specialty probes, expert power measurement software, logic analysers, signal generators and real-time spectrum analysers are among the general-purpose equipment used in automotive design as well as production. "Switch-mode power supplies, electronic loads and data loggers are used in design and verification and life-cycle testing of any automotive component or system," shares Prabhakaran.

"Equipment can be standalone box tester units with predefined test functionality or modular with I/O cards added based on the sub-system being tested," says Rohit Panikkar, technical marketing manager, National Instruments. During R&D, dynamometers are used to study the effect of various conditions on engine, brake and chassis. Test systems to study the endurance of system components

and noise, vibration and harshness (NVH), as well as safety test systems are also used at this stage. An RF signal generator and RF analyser with spectrum analysis capability are used to test remote keyless-entry modules.

Automotive vehicle power systems are very harsh electrical environments. "High-current motors, solenoids and other components on the power system often cause power system voltage transients and dropouts to occur. Successful operation of automotive electronics depends on adequate power transient immunity. These transients can be studied using a scope. However, a DC power analyser reduces the test time and improves ECU test coverage by allowing engineers to easily capture and re-play power system transients. Libraries of transient cases can be developed and archived for future use as well as for long-term support and ECU re-qualification," explains Jain.

Data acquisition systems and highly accurate digital multimeters (DMMs) are also required in almost all the stages of design and development cycles. "There are specially designed loads and power supplies used for

testing of fuel cells, batteries and charging of electrical vehicles," adds Prabhakaran.

Automated test systems

In automobile production, typically, automated test equipment (ATE) are used to achieve high throughput and efficiency. ATE are built using COTS equipment like oscilloscopes, spectrum analysers, DC power analysers, data acquisition systems and switches with a difference that these are interconnected via LAN or GPIB and the software runs the required test routines and hence controls these equipment. A lot of T&M vendors provide modular or synthetic instrumentation, which can be combined to form an ATE depending on the specific requirement.

Jain shares that there are turn-key test solutions for engine control module (ECM) testing which are primarily used in a manufacturing set-up where time (test development, execution and system deployment), cost (capital and integration) and scope (throughput, accuracy and flexibility) are of utmost importance.

"T&M vendors also provide solutions based on the OpenChoice structure, which means customers can easily develop their own software or utilities script for an automation testing mechanism," adds Sun. Also, "a small number of manufacturers use innovative processes by transmutation of communication technology for verification of mechanical parameters," says Gautam.



EMI test receiver from Rohde & Schwarz